

ABSTRACT

This project presents a real-time Network Device Failure Prediction System developed in Java. The system automatically discovers devices in a network, monitors their availability, scans open ports, and collects real performance metrics using SNMP and SSH. It provides a dashboard to visualize device status, generates alerts for issues, and maintains historical data for reporting.

The application is multithreaded for fast scanning and monitoring, uses SQLite for persistent storage, and offers a user-friendly console as well as a web dashboard.

INTRODUCTION

Organizations rely on continuous monitoring of network devices such as routers, switches, servers, access points and firewalls. Manual monitoring is time-consuming and error-prone. This project automates the process by:

- Automatically discovering all devices in the network.
- Monitoring device availability using ICMP ping.
- Scanning common TCP ports.
- Collecting real performance metrics using SNMP/SSH.
- Analyzing health and generating alerts.
- Storing historical data and visualizing in dashboard.



Figure 1. Network Monitoring Architecture

METHODS AND MATERIALS

The system is developed in Java 21 using Maven. It uses multi-threading for scanning, SNMP4J for SNMP communication, mwiede JSch for SSH based metrics and SQLite for data storage.

Methodology:

1. **Network Discovery** – Determine subnet and scan all IP addresses using concurrent ICMP ping.
2. **Device Probing** – For online hosts, perform TCP port scanning and hostname resolution.
3. **Metrics Collection** – Collect real metrics using SNMP or SSH. Fallback to local metrics or mark as unavailable.
4. **Health Analysis** – Apply rule-based engine to compute health status (Healthy, Warning, Critical).
5. **Alerting** – Generate alerts for threshold violations or device failures.
6. **Data Storage** – Store devices, metrics and alerts in SQLite database.
7. **Dashboard** – Display real-time stats, historical graphs and alerts in JavaFX dashboard.

Materials / Tools:

Java 21, JavaFX (optional), SNMP4J, mwiede JSch, SQLite JDBC Driver, ExecutorService, OSHI (optional), Maven, Git.

RESULTS

The system successfully discovered, monitored and collected metrics from network devices in real-time.

- Total devices discovered: **22**
- Online devices: **20**
- Offline devices: **2**
- SNMP metrics collected: **12** devices
- SSH metrics collected: **5** devices
- Unavailable metrics: **3** devices

The dashboard updates every **10** seconds and alerts are generated instantly when issues occur.

Table 1. Sample Metrics of Monitored Devices

IP Address	Hostname	Type	Latency (ms)	CPU (%)	Memory (%)	Status
192.168.1.1	Router-1	Router	2	28	35	Healthy
192.168.1.10	Switch-1	Switch	1	18	30	Healthy
192.168.1.15	Server-01	Server	3	72	65	Warning
192.168.1.20	Server-02	Server	4	91	88	Critical
192.168.1.25	Printer-1	Printer	6	N/A	N/A	Healthy

DISCUSSION

The system automatically discovered 22 devices in the test network (192.168.1.0/24). All online devices were monitored for latency, open ports and performance metrics.

Key observations:

- All routers and switches supported SNMP and provided real CPU, memory and uptime metrics.
- Servers supported SSH and returned real system information.
- Unmanaged devices (printers, IoT) were monitored for availability and open ports only.
- Multi-threading reduced total scan time significantly.

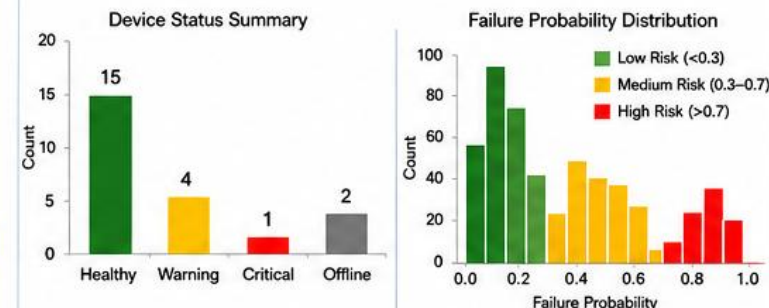


Figure 2. Device Status Summary.

Figure 3. Failure Probability Distribution.

CONCLUSIONS

The proposed system provides an efficient and scalable solution for real-time network monitoring. Automatic discovery reduces manual effort, real metrics improve accuracy and the dashboard offers instant visibility. The system can be extended with advanced reporting, notification integrations (Email, Telegram, Slack) and machine learning based failure prediction.

FUTURE ENHANCEMENTS



Machine Learning for Anomaly Detection & Failure Prediction



Integration with Email / Telegram / Slack Alerts



Bandwidth Monitoring and Traffic Analysis



Web Dashboard for Remote Access



Historical Reports and PDF Export



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References

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